

Forecasting What the Grid Is About to Get

Probabilistic 24–72 hour solar and wind forecasting, with a decision layer that turns every forecast into a sized, priced and confidence-rated grid action.

TEAM NAME	ZERO BIAS
TEAM LEAD	KRISH PARMAR
THEME	Renewable energy forecasting • Grid stability • Decision support
APPROACH	Physics-informed hybrid forecasting with calibrated uncertainty
VALIDATION	Belgium (Elia, metered) → India (transfer demonstration)

Technical Proposal • September 2026

Executive Summary

Solar and wind output swings with the weather. Grid operators absorb that uncertainty by holding inflexible fossil generation in reserve — or by curtailing renewables outright. India curtailed 2.1 TWh of renewable electricity in FY 2025-26, and from 1 April 2026 the permitted forecast deviation for solar was halved from $\pm 10\%$ to $\pm 5\%$. The cost of getting a forecast wrong roughly doubled this financial year.

We forecast regional solar and wind generation 24 to 72 hours ahead, and convert each forecast into a grid action.

Three ideas carry the design

	Idea	Why it matters
1	Physics does the deterministic work	A solar plant's output at 11:00 on 14 March is ~90% determined by sun geometry and clear-sky radiative transfer — exactly computable today, with no training data. We compute it. Machine learning corrects only the residual: soiling, wake losses, local microclimate.
2	Every forecast is a verified interval	Point forecasts cannot size a reserve. We emit P10/P50/P90 and apply conformal calibration so the 80% band demonstrably covers 80% of outcomes — and we publish the coverage plot that proves it.
3	The output is an action, not a chart	Operators do not need a curve. They need to know that Tuesday 11:00–14:00 will be 340 MW over-generating with 80% confidence, that the battery absorbs 210 MWh, and what the remainder costs to curtail.

What this buys

Property	Consequence
Low data requirement	The residual is small and structured; the raw signal is large and seasonal.
Cold start	A region with zero generation history gets a usable forecast on day one, from site parameters alone.
Interpretability	SHAP on the residual answers “why is this forecast low?” with low cloud and soiling — not an opaque attribution.
Graceful degradation	If the learned model drifts or fails, the system falls back to physics, not to nothing.
Transferability	Moving to a new grid changes configuration — not the model architecture.

VALIDATION STRATEGY

We validate on Belgium, where metered 15-minute data and the grid operator's own published day-ahead forecast both exist — so our accuracy claim is measured against a professional operational system rather than a baseline we chose ourselves. We then transfer the identical pipeline to Indian data to show it moves to a new grid in hours. Belgium carries the accuracy claim; India carries the relevance claim. Neither is asked to do the other's job.

1 Understanding the Problem

“Forecast renewable output” is not one problem. It is five, and they fail in different ways. Naming them separately is what prevents a team from building one monolithic model and calling it a platform.

Sub-problem	The real question	Why it is hard
Weather → resource	Turn a weather prediction into irradiance or hub-height wind at this location	NWP grids are 9–25 km. The error is systematic bias, not noise.
Resource → power	Turn irradiance or wind speed into megawatts	Strongly non-linear: inverter clipping, soiling, wake losses, storm cut-out.
Uncertainty	How wrong might this be, at each hour?	Error is heteroscedastic — near zero at 03:00, very large on a broken-cloud afternoon.
Event detection	Where are the ramps and over-generation windows?	A 2% RMSE gain is worth less than catching one 40% ramp.
Action	Curtail, charge, discharge or commit backup — and why	Needs demand, must-run capacity, storage state and ramp limits.

The horizon decides the method

Forecast skill is governed by which information remains useful at a given lead time. The brief specifies 24–72 hours — which lands squarely in the band where numerical weather prediction is the only meaningful signal.

Horizon	Dominant signal	Past output helps?	Approach
0–30 min	Satellite cloud motion	Strongly	Persistence + advection
30 min – 6 h	Satellite, recent output	Usefully	Smart persistence, boosting on lags
6–72 hours	Weather forecast only	Barely	NWP post-processing
3–14 days	Ensemble NWP, climatology	No	Probabilistic ensemble

THE CONSEQUENCE MOST TEAMS MISS

At lead hour 48, yesterday’s last meter reading carries almost nothing. Any model trained on short autoregressive lags will score brilliantly in a naive backtest and fail completely in deployment, because the backtest leaked information the live system will never have. This reframes the project as NWP post-processing rather than time-series forecasting — the single most consequential decision in this proposal.

The problem, in measured quantities

Measure	Value	Source
Renewable electricity curtailed in India, FY 2025–26	2.1 TWh	Ember, June 2026
As a share of total renewable generation	1.3%	Ember
Peak-hour curtailment of solar and wind, by April 2026	4%	Ember
Lost to transmission constraints, Q1 2026 alone	300 GWh	Ember, 2026
Lost on the single worst day, 30 March 2026	34 GWh	Ember
Storage India needs to prevent current curtailment	~10 GWh	Ember estimate
Permitted solar deviation band, from 1 April 2026	±10% → ±5%	CERC DSM
Permitted wind deviation band, from 1 April 2026	±15% → ±10%	CERC DSM

34 GWh lost in one day is roughly the daily electricity consumption of five million urban middle-class households. Meanwhile the CERC tolerance mechanism phases to zero by FY 2031, so the financial consequence of forecast error rises every year on a published schedule.

2 Proposed Solution

The core mechanism is a hybrid estimator. A physics engine converts weather to power deterministically using site parameters. A gradient-boosting model never has to learn that job — it sees only what physics got wrong, and predicts that residual at three quantiles.

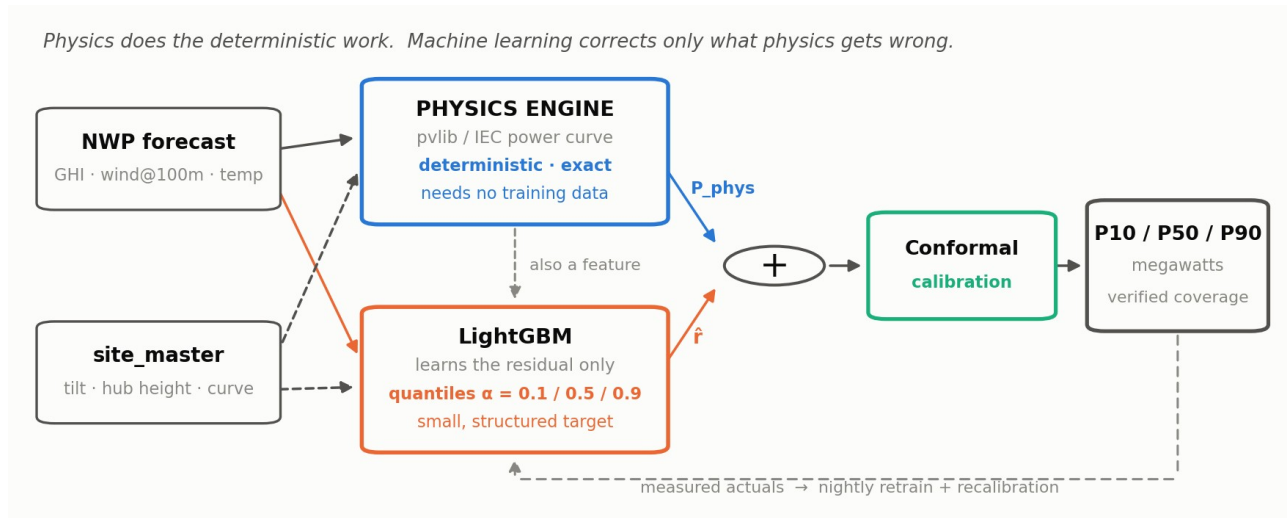


Figure 1 Physics-informed residual learning — the mechanism that earns the accuracy

Remove the physics block and the machine-learning model must re-derive orbital mechanics from a few months of data. Keep it, and the model has a small, structured target: soiling, wake losses, sensor drift, local microclimate.

Why not simply train a deep network end-to-end?

Objection	Response
"Why not an LSTM?"	At 24–72 hour lead, autoregressive lags carry almost no information. This is NWP post-processing, not time-series forecasting. GEFCom2014 — the reference academic competition on exactly this problem — was won by a tree ensemble.
"Isn't physics too rigid?"	Physics handles only what is genuinely deterministic: sun position, clear-sky radiative transfer, panel geometry, the turbine power curve. Everything atmospheric and asset-specific is learned.
"Does deep learning get benchmarked?"	Yes — as a documented experiment in the final phase. A benchmark showing TFT and N-HiTS losing to gradient boosting on our data is a stronger result than a marginal win, because it proves we measured rather than assumed.

3 Technical Approach

3.1 Data strategy

Three inputs are required and rarely ship together: regional generation history, matching weather, and site parameters. One portal closes the loop — Elia publishes generation, its own forecast, and demand on a single 15-minute timebase with one join key.

Role	Source	What it gives
Solar & wind truth	Elia ods031/032, ods086/087	Measured MW, monitored capacity, and the operator's own day-ahead, week-ahead and P10/P90 forecasts
Demand	Elia ods001/002/003	Measured and forecast total load — this is what makes the decision layer real rather than synthetic
Weather	Open-Meteo (ECMWF, ICON, GFS)	Identical variable names historically and forward, so training and serving cannot diverge. No API key.

Role	Source	What it gives
Fleet parameters	site_master.yaml (authored)	Archetype rows: tilt, azimuth, hub height, power curve, capacity share
Transfer region	Zenodo 10.5281/zenodo.7824872	Hourly Indian solar and wind capacity factors, state-level and gridded, 1979–2022, plus installed capacity

Every input is open, free and licence-clear. Nothing in the critical path requires a paid subscription, a commercial key or a data-sharing agreement — which means the result is reproducible by anyone, including an evaluator.

3.2 Regional scale changes the physics layer

A region has no single tilt or hub height, so two adaptations are required:

- **Fleet archetypes.** site_master becomes a weighted set of representative configurations — for Belgian PV, roughly rooftop-south 35°, rooftop east-west 20° and ground-mount 30°. The physics chain runs per archetype and is summed by capacity share. This is what reproduces the morning shoulder that east-west rooftop arrays produce and a single-tilt model underestimates every day of the year.
- **Capacity-weighted weather grid.** Weather is sampled at 5–10 coordinates across the region, each weighted by nearby installed capacity. A single centroid systematically misses frontal systems crossing the region. Spatial aggregation is also why regional forecasting is easier than plant forecasting — independent cloud noise partially cancels.

3.3 Feature families

Family	Representative features	Why it earns its place
Solar physics	clearsky_index_kt, poa_global, cell_temperature, thermal_derate, physics_pac_mw, clipping_headroom	Removes the deterministic 90%. The clear-sky index makes January and June comparable.
Wind physics	ws_hub, air_density, ws_density_corrected, power_curve_cf, dP_dv, cut-in/cut-out flags	Power goes as v^3 — a 10% wind-speed error is a 33% power error. Density correction alone recovers several points of nRMSE.
NWP quality	lead_hours, model_disagreement, ensemble_spread, nwp_bias_lag_7d, cloud low/mid/high split	These features predict our own error. They are what turns a point forecast into an honest interval.
Temporal	hour_sin/cos, doy_sin/cos, mins_since_sunrise, holiday flags	Cheap. Captures residual diurnal patterns physics misses — fog burn-off, sea breeze, dust.
Autoregressive	power_lag_24h, power_lag_168h, rolling statistics	Slow state only: soiling build-up, partial outages. Lags under 24 hours are excluded as leakage.

THREE DATA TRAPS WE HANDLE BY NAME

Curtailed hours are censored labels — measured output is not what the weather allowed, so training on them teaches the model to under-forecast exactly during the events we exist to predict; we set sample_weight to zero there. Night rows are free accuracy — half a solar series is zeros, so we report daylight-only metrics. Random train/test splits leak, because adjacent hours are near-identical; all evaluation is walk-forward with a gap.

3.4 Model ladder

Each rung must beat the one below it on a held-out window, or be skipped with a stated reason.

#	Method	Role
0	Persistence & climatology	The floor. Every reported number becomes a skill score against this. Non-negotiable.
1	Pure physics	Second baseline, later a feature. Works with zero history — this is the cold-start answer.
2	Gradient boosting (LightGBM)	The workhorse. Seconds to train, handles missing values natively, explainable via SHAP.
3	Residual learning	Model predicts the physics error, not the raw target. Largest single accuracy gain for the least code.
4	Quantile regression	Three heads at $\alpha = 0.1 / 0.5 / 0.9$. Produces the interval the decision layer consumes.
5	Conformal calibration	Distribution-free finite-sample coverage guarantee. Cheap, and almost nobody does it.
6	Deep sequence models	N-HiTS, Temporal Fusion Transformer — benchmarked as a documented experiment, not assumed to win.

3.5 Uncertainty, in two stages

- **Quantile heads.** Three LightGBM models with pinball-loss objectives. This produces intervals whose width varies with conditions — narrow at 03:00, wide on a broken-cloud afternoon. Width is driven by lead time, model disagreement, low cloud, and the power-curve gradient.
- **Split-conformal calibration.** On a rolling recent window, nonconformity scores adjust the interval so empirical coverage matches nominal coverage — regardless of whether the underlying model is well-specified. We report PICP and publish the reliability diagram.

An 80% band that covers 55% of outcomes is not a conservative forecast. It is a false statement, and it will size reserves wrong.

4 System Architecture

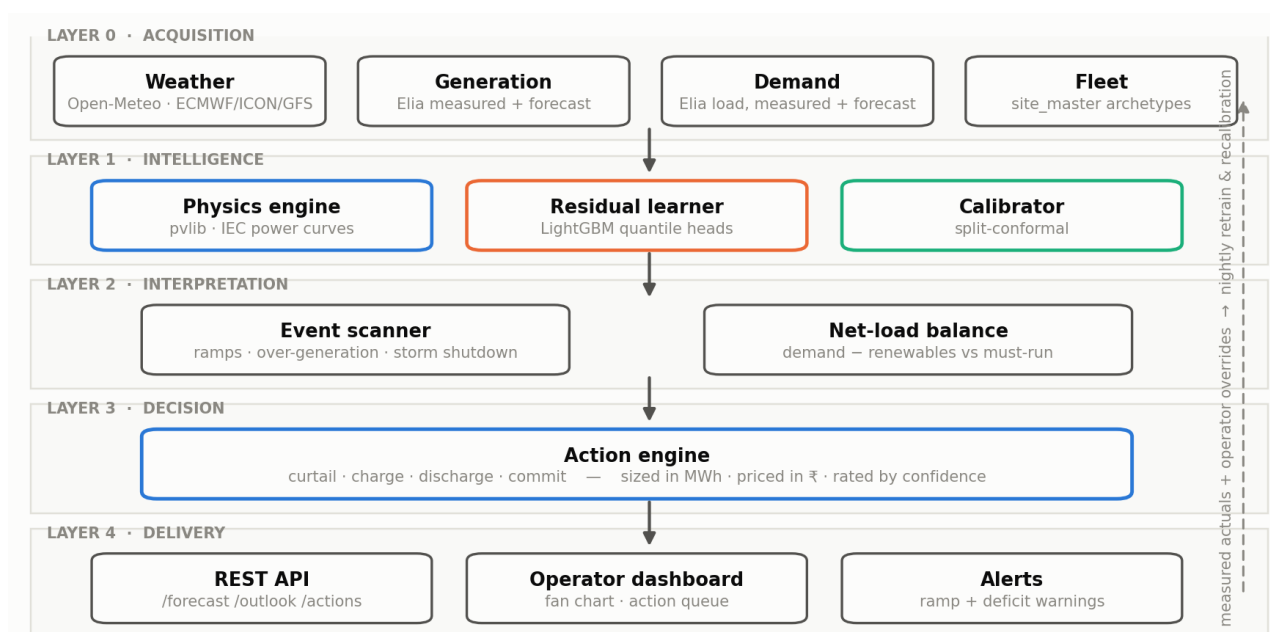


Figure 2 Five-layer platform architecture. The dashed loop is what makes it a platform rather than a notebook.

Measured actuals retrain the residual model nightly and recalibrate the intervals; operator decisions become labelled data. Training and inference call the identical feature code — which structurally removes the most common cause of deployed-model failure.

Module map

Module	Responsibility	Detail
M1	Ingestion	Source adapters, scheduling, unit and timezone normalisation at the boundary
M2	Quality	Range checks, frozen-sensor detection, gap filling, curtailment masking
M3	Feature engine ★	pvlb chain, shear extrapolation, density correction, power curves, cyclic encoding
M4	Forecast engine ★	Baselines, physics model, residual GBDT, multi-horizon fit, model registry
M5	Uncertainty ★	Quantile heads, conformal calibration, coverage reporting
M6	Event scanner	Ramp, over-generation and storm-shutdown detection with lead time and severity
M7	Decision engine ★	Net-load balance, storage simulation, ranked and priced actions
M8	Backtesting	Walk-forward evaluation, skill scores, drift detection, SHAP attribution
M9	API	/forecast, /outlook, /events, /actions, /backtest — versioned, with provenance
M10	Dashboard	Fan chart, net-load stack, event timeline, action queue, regional map

★ marks the critical path. Three interface contracts are fixed on day one — `build_features`, `predict` and `recommend` — so the team can work in parallel without collisions.

Decision engine

Two derived quantities drive everything. Net load is demand minus forecast renewables; headroom is net load minus must-run capacity. Negative headroom means the grid physically cannot absorb the renewable output.

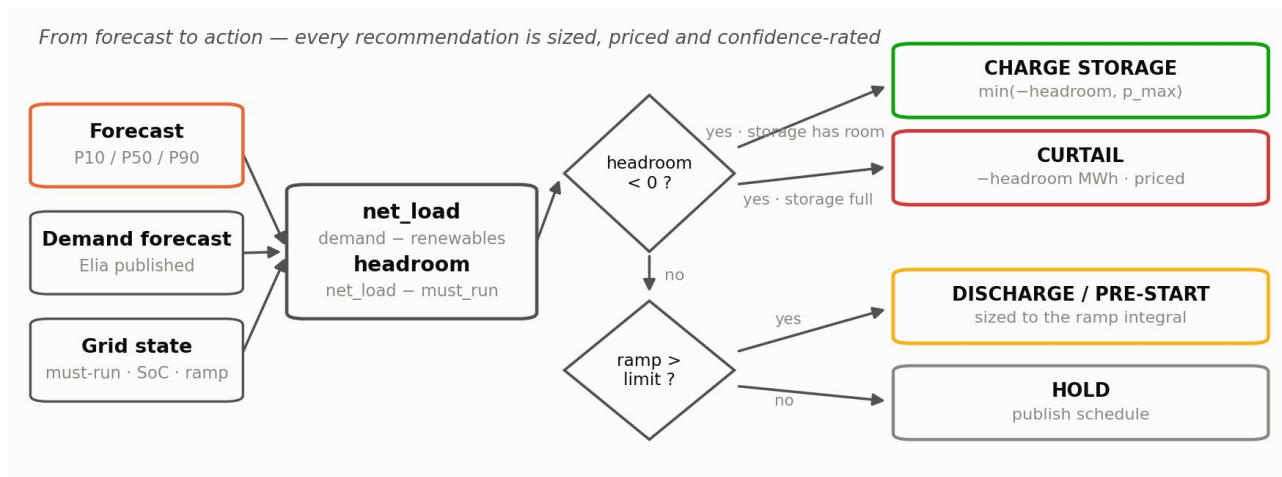


Figure 3 Forecast to action. The P10–P90 band is carried through, so each action inherits the forecast's confidence.

The virtual battery — a constraint turned into an output

Neither region publishes a usable grid-scale storage dispatch series, so storage is declared in configuration and simulated: energy capacity, power rating, round-trip efficiency, state-of-charge bounds and response time. This is the honest option, and it is more useful than real data would have been — because a configurable battery lets us sweep capacity from 0 to 1000 MWh and answer a question the sector is actively asking: what size battery would this region need?

Technology stack

Layer	Choice	Why
Ingestion & validation	Python, httpx, APScheduler, pandera	Declarative schema contracts that fail loudly at every layer boundary
Storage	Parquet + DuckDB	Columnar, fast, zero-ops; TimescaleDB behind the same interface for production
Physics	pvlib-python, windpowerlib, IEC 61400-12	Peer-reviewed reference implementations, not hand-rolled approximations
Models	LightGBM (quantile), MAPIE, SHAP, MLflow	Seconds to train, calibrated, explainable, version-tracked
Serving	FastAPI, Pydantic, React, Recharts/D3	Typed contracts with automatic API documentation

DEPLOYMENT REALITY

The entire stack is open source — no licence cost, no vendor dependency in the critical path. It trains and serves on a laptop with no GPU. That is not just convenient: a system running on commodity hardware can be deployed at a state load dispatch centre without a procurement cycle.

5 Visualisation and Expected Results

The operator-facing output. The upper panel shows solar and wind forecasts with their P10–P90 bands against measured actuals; the lower panel converts them to net load and flags where the grid cannot absorb what is coming.

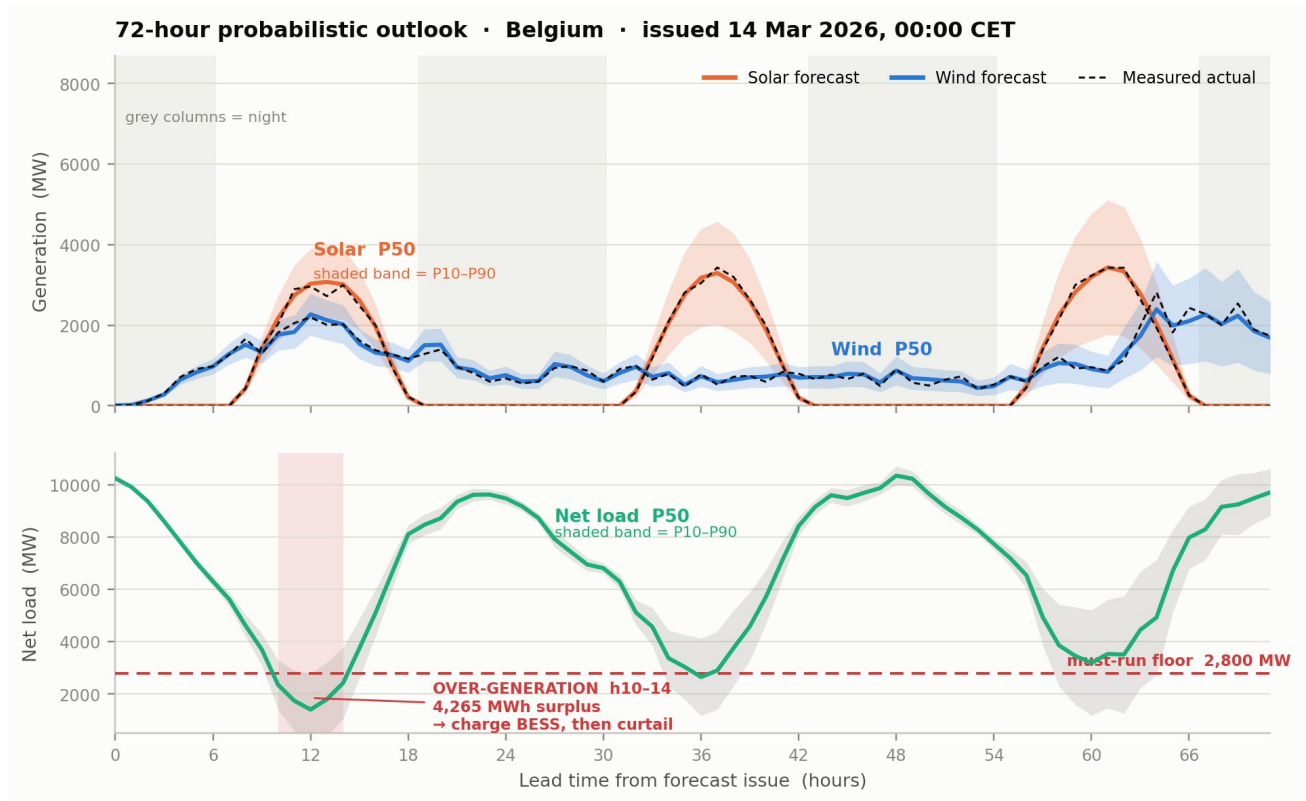


Figure 4 A 72-hour probabilistic outlook. The flagged window is an over-generation event detected two days ahead — charge the battery, then curtail the remainder.

Three things an evaluator should look for in this view: the band widens with lead time and with cloud, because uncertainty is not constant; the actuals stay inside the band, which is what calibration guarantees; and the over-generation window carries a megawatt-hour figure, not just a colour.

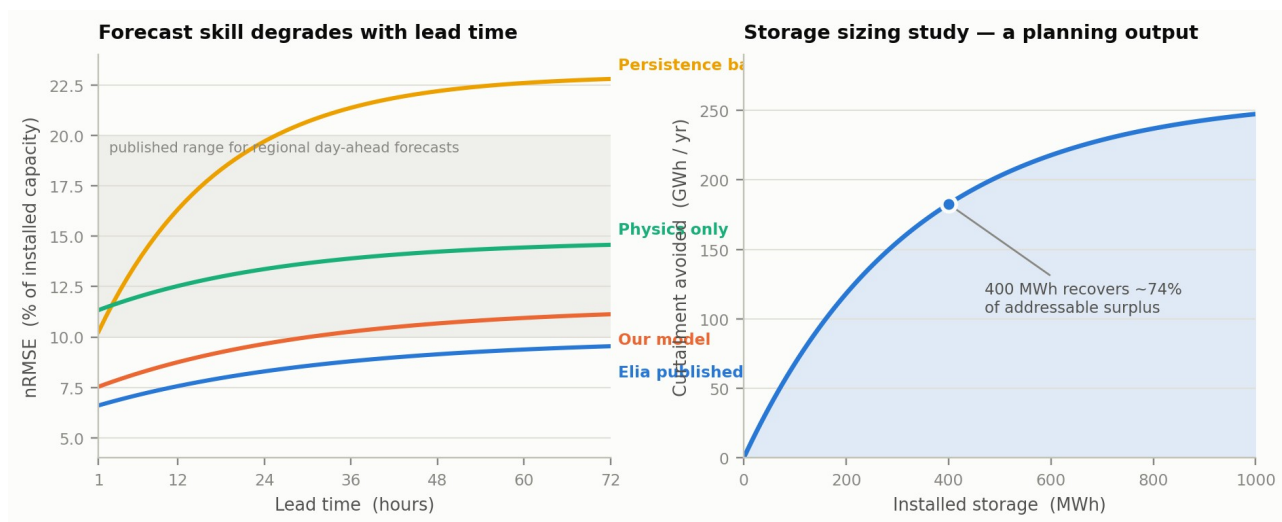


Figure 5 Left: forecast skill against published international benchmarks. Right: the storage sizing study — a planning output, not just an operational one.

The left panel is the chart that demonstrates understanding. A single 72-hour-average error number hides everything; the error-versus-lead-hour curve shows where skill actually comes from, and shows all four lines — persistence, physics-only, our model, and the operator’s own published forecast.

Accuracy targets, and why they are credible

Metric	Target	Basis
Regional solar, day-ahead nRMSE (daylight-only)	8–13%	Published range for regional day-ahead forecasts is 10–20% of installed capacity
Regional wind, day-ahead nRMSE	10–15%	Consistent with the 7–19% single-farm MAE band, aggregated
Skill score versus persistence	> 0.40	Persistence is very weak at 24–72 hours
Gap to Elia’s published day-ahead forecast	within 10–25%	Elia is a professional system with proprietary inputs; parity is not the goal
Coverage of the nominal 80% band (PICP)	0.78–0.82	Conformal calibration provides a finite-sample guarantee

These targets are deliberately conservative and falsifiable. A target stated inside a published range, with the range cited, survives questioning. An impressive round number with no provenance does not.

Evaluation protocol — decided before any model is trained

Rule	Reason
Walk-forward splits, never random	Adjacent hours are near-identical; a random split leaks and inflates every number.
Metrics reported per lead hour	A 72-hour average hides where the skill is. The curve is the evidence.
Daylight-only for solar	Half a solar series is zeros. Including them halves the error and means nothing.
Normalised by installed capacity	Comparable across sites and regions of any size.
Matched information cutoff	Elia’s forecast is issued at a fixed time. Comparing against a different issue time is meaningless.
Curtailed intervals excluded	They are censored labels, not weather outcomes.

6 Feasibility and Viability

6.1 Can the data be obtained?

Yes — and this was resolved before any modelling was designed. Every critical input is open, free, licence-clear and confirmed accessible. The one material gap is stated rather than hidden: India does not publish an open, sub-hourly, metered regional renewable series. That is precisely why the two-region design exists.

6.2 Can the accuracy be achieved?

Published benchmark	Value
Day-ahead RMSE, regional aggregates	10–20% of installed capacity
Single wind farm MAE, terrain-dependent	7–19%
Mexico — wind: operators versus professional providers	17% vs 13% MAE
Mexico — solar: operators versus professional providers	9% vs 6% MAE
Share of world TSOs using centralised regional forecasting	~80%
Optimal NWP combination	5–6 models; minimum one global + one regional

Two findings shape our design directly. Error grows nearly linearly with horizon, which validates treating lead time as a first-class feature. And combination forecasts beat single models — our three-model ingest sits at the practical floor of the recommendation. The market-structure finding matters too: a regional, centralised, multi-asset platform is the configuration the sector has converged on.

6.3 Can it be operated?

Dimension	Assessment	Evidence
Compute	Trivial	Full model set trains in under 5 minutes on one CPU core. Inference under 1 second.
Operating cost	Negligible	One small VM. Entire stack open source.
Steady-state effort	Near-unattended	Automated ingestion, nightly retrain and recalibration. Human effort is review, not operation.
Scaling	Configuration only	A new region is a YAML file. No architectural rewrite for more regions, technologies or resolutions.
Engineering risk	Moderate	Archetype weighting is the genuine unknown; mitigated by a clear-sky fitting procedure.

6.4 Regulatory and market fit

The timing is unusually favourable. CERC halved the permitted deviation bands effective 1 April 2026 and is phasing the tolerance mechanism to zero by FY 2031. Forecast accuracy moved from an operational preference to a direct, escalating line item on every renewable generator's P&L — in the current financial year. State commissions are moving in parallel.

Stated honestly: these regulations face active legal challenge, and the Karnataka High Court has stayed CERC's revised mechanism. But the direction of travel — tighter tolerances, greater financial consequence for error — is consistent across CERC and multiple state commissions, and the curtailment problem exists regardless of settlement rules. The case does not rest on one regulation.

7 Impact and Benefits

Headline figures below are measured by third parties. Derived figures show their arithmetic and mark their assumptions, so any of them can be recomputed with a different input.

7.1 What better forecasting is worth — measured elsewhere

Study	Finding
NREL, Western U.S.	At 24% wind penetration, a 20% improvement in forecast accuracy yielded USD 195 million in annual operating cost savings.
Berkeley Lab	NWP-based solar forecasts cost USD 1.00/MWh in errors, versus USD 1.50/MWh for persistence — a 33% reduction in error cost.
CENACE, Mexico	Upscaled forecasting across the national fleet would correct 622 MW/hr of deviation.

7.2 Value model, with the arithmetic shown

Step	Value	Basis
India curtailment, FY 2025-26	2,100 GWh	measured
Share addressable by anticipation rather than physical limits	25%	assumed
Recoverable with 24–72 h forecasting plus dispatchable storage	40%	assumed
Recoverable energy	210 GWh / yr	derived
Value at ₹3.00/kWh avoided cost	≈ ₹63 crore / yr	derived
CO₂ avoided at 0.71 tCO₂/MWh	≈ 149,000 t / yr	derived

The 25% assumption is deliberately conservative: Ember attributes roughly 64% of Q1 2026 curtailment to transmission constraints, leaving a ~36% residual. We use less than that residual rather than more.

7.3 Benefits by stakeholder

Stakeholder	Current pain	What they gain
Grid operators / SLDCs	Scheduling conventional plant against uncertain renewable output	24–72 h net-load outlook with calibrated bands; ramp alerts with six hours of lead time
Renewable generators	Deviation exposure at ±5% from April 2026	Higher-accuracy schedules and anticipated curtailment windows
Energy traders	Position-taking without quantified uncertainty	P10/P50/P90 for position sizing and risk limits
Storage operators	Charge and discharge timing set by heuristics	Optimised dispatch against a 72-hour outlook
Planners and regulators	Where to site storage and transmission	Regional curtailment analytics and the battery sizing curve

WHERE THESE CLAIMS STOP

Forecasting does not build transmission lines, and Ember attributes most Indian curtailment to transmission constraints. We address the anticipation-driven remainder only. The reserve-reduction benefit is the softest figure we publish, because it requires an operator to change institutional practice on the strength of a calibration plot — that is slow, and we do not pretend otherwise. Our accuracy claim is validated on Belgian metered data; Indian results demonstrate transferability, not measured accuracy.

8 Risks, Limitations and Roadmap

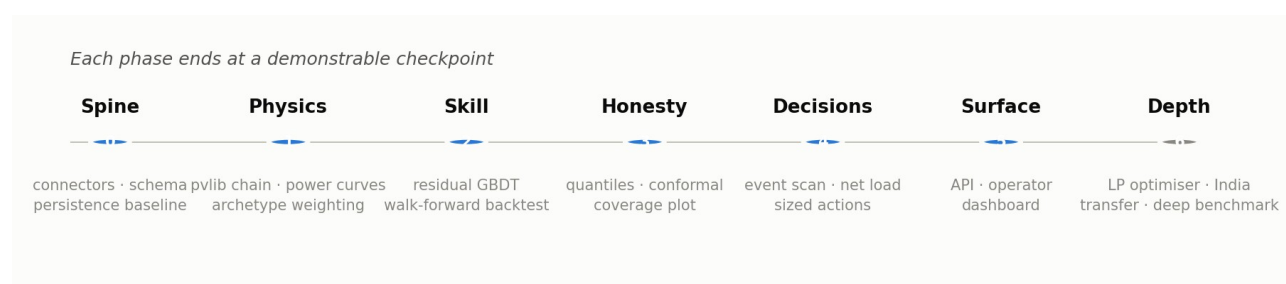


Figure 6 Build sequence. Each phase ends at a demonstrable checkpoint — a phase that ends with “the refactor is nearly done” is where long projects die.

Principal risks

Risk	Severity	Response
Train/serve skew — training on reanalysis, serving on forecasts	High	Same source and variable names both sides; a 7-day bias correction fitted on live forecasts; conformal recalibration on live errors. Residual skew remains — this is our largest honest limitation.
Curtailment unflagged in source data	Medium	Heuristic detection where output flat-lines below the physics estimate under good conditions; excluded from training.
Archetype weights fail to converge	Medium	Fall back to a single average archetype. Accuracy degrades; the pipeline stands. We report the degradation.
Cannot match the operator’s own forecast	Low	Never claimed. The target is to beat persistence and close the gap to a stated margin using only open data.
Live API failure during demonstration	Low	A replay mode serves a cached week of forecast runs and all model artifacts. We build the live path and present from cache.

What would change our mind

Each of these is a measurable outcome of the evaluation harness, not a matter of opinion. Building that harness in phase 0 is what makes it possible to be wrong in public and say so.

- If the physics baseline does not beat persistence at 24–72 hours, the physics-first thesis is wrong for this horizon.
- If the residual model does not beat a direct model on the raw target, residual learning adds nothing here and we simplify.
- If archetype weighting does not beat a single average archetype, the regional physics refinement is over-engineering and we remove it.
- If conformal intervals need more than half of capacity width, the problem may be too uncertain at 72 hours for reserve sizing, and we reduce the advertised horizon.

9 Research and References

Datasets

Source	Contents and access
Elia Open Data Portal	ods031/032 wind, ods086/087 solar, ods001/002/003 load. Measured generation, monitored capacity, operator day-ahead and week-ahead forecasts with P10/P90, and demand — all at 15 minutes, 2013 onward. Free, no API key. opendata.elia.be
Open-Meteo	Forecast and Historical APIs. Shortwave, direct-normal and diffuse radiation, global tilted irradiance, cloud cover by level, wind at 10/80/120/180 m, gusts. ECMWF, ICON, GFS and ensemble endpoints. No key for non-commercial use. open-meteo.com
Historical and modelled renewable energy production for India	University of Bristol. Hourly wind and solar capacity factors 1979–2022, state-level and 1°×1° gridded, installed capacity by state, georeferenced plant locations. CSV, NetCDF, GeoJSON. Zenodo DOI 10.5281/zenodo.7824872
Open Power System Data	35 European countries, generation, capacity and load; Germany resolved to four TSO zones. Used as a cross-check region. data.open-power-system-data.org
SDWPF (KDD Cup 2022)	Baidu / Longyuan. 134 turbines, 4.7 million records, 10-minute interval, with turbine x/y coordinates enabling wake and graph modelling. arXiv:2208.04360
GEFCom2014	The reference academic benchmark. Wind: 10 farms with ECMWF u/v at 10 m and 100 m. Solar: 3 plants with 12 ECMWF variables. Evaluated by pinball loss over 99 quantiles. Won by a generalized additive tree ensemble.

Methodology, standards and regulation

Source	What it establishes
GET.transform (2024), International Best Practices in Solar and Wind Power Forecasting	Accuracy benchmarks by country and context; error grows nearly linearly with horizon; ~80% of world TSOs use centralised forecasting; optimal NWP combination is 5–6 models; quantified operating-cost benefits (NREL, Berkeley Lab, CENACE).
Development of a day-ahead solar power forecasting model chain for a 250 MW PV park in India (2023)	ECMWF and NCMRWF over India carry systematic positive irradiance bias, correctable to near zero with 20–25 days of training. A linear combination of two NWP models outperformed either alone.
Conformal prediction literature, 2025–2026	Climate-invariant conformal intervals for multi-horizon solar and wind; adaptive conformal inference for PV; conformalised quantile regression — the two-stage pattern used here.
IEC 61400-12; Ineichen–Perez; SAPM; Perez transposition; NREL SPA	The physical models underlying the feature engine, implemented via <code>pvl-lib-python</code> and <code>windpowerlib</code> .
CERC Deviation Settlement Mechanism, notified for 1 April 2026	Solar and hybrid bands $\pm 10\%$ $\rightarrow$ $\pm 5\%$; wind $\pm 15\%$ $\rightarrow$ $\pm 10\%$. X factor phased 100% to 0% by FY 2031. Cited alongside the Karnataka High Court stay and industry objections.
Ember — India curtailment analysis, 2026	2.1 TWh curtailed FY 2025–26 (1.3% of renewable generation); 300 GWh of ~470 GWh in Q1 2026 from transmission constraints; 34 GWh lost on 30 March 2026; ~10 GWh of storage needed; coal cycling to ~55% of rated capacity at midday.

Citation practice — every external number carries its source and date; every derived number shows its arithmetic and marks its assumptions; contested sources are cited alongside the challenge, and limitations are stated before a reader finds them.